1. A particle moves along a straight line. Its displacement S varies with time t according to the law $S^2 = at^2 + 2bt + c$ (a, b and c are constants). The acceleration of this particle varies as

   A. S^0
   B. S^{-1}
   C. S^{-2}
   D. S^{-3}

Answer (D)

Sol. $S^2 = at^2 + 2bt + c$

Differentiate w.r.t. time, we get

$$2S\frac{dS}{dt} = 2\,at + 2b$$

$$s\frac{dS}{dt} = at + b \Rightarrow \frac{dS}{dt} = \frac{at+b}{S}$$

$$\left(\frac{dS}{dt} = v\right)$$

Again, differentiate w.r.t. time, we get

$$\frac{dS}{dt}\left(\frac{dS}{dt}\right) + S\left(\frac{d^2S}{dt^2}\right) = a$$

$$S\left(\frac{d^2S}{dt^2}\right) = a - \left(\frac{dS}{dt}\right)^2$$

$$S\left(\frac{d^2S}{dt^2}\right) = a - \left(\frac{at+b}{S}\right)^2$$

$$S\left(\frac{d^2S}{dt^2}\right) = \frac{aS^2-(at+b)^2}{S^2}$$

$$S\left(\frac{d^2S}{dt^2}\right) = \frac{a\left(at^2+2bt+c\right)-\left(a^2t^2+b^2+2\,abt\right)}{S^2}$$

$$\frac{d^2S}{dt^2} = \frac{ac-b^2}{S^3}$$

$$i.e.\ Acc. \propto S^{-3}$$

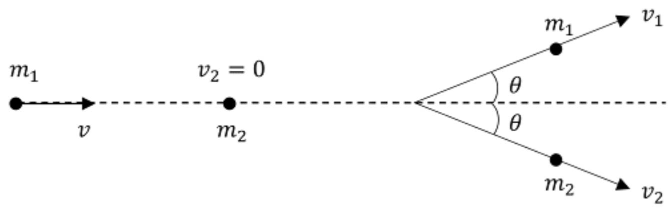
2. A ball A (mass m_1) moving with velocity v experiences an elastic collision with another stationary ball B (mass m_2). Each ball flies apart symmetrically relative to the initial direction of motion of ball A, at an angle θ.

Ratio of the masses of balls $\frac{m_1}{m_2}$ *is*

A. 1 + 2 cosθ
B. 2 cos 2θ
C. 1 + 2 cos 2θ
D. 1 + cos 2θ

Answer (C)

Sol.



In vertical direction ⇒ $m_1 v_1 \sin sin\,\theta = m_2 v_2 \sin sin\,\theta$

$\therefore m_1 v_1 = m_2 v_2$

$\therefore \frac{m_1}{m_2} = \frac{v_2}{v_1} \ldots (1)$

In horizontal direction ⇒

$m_1 v + 0 = m_1 v_1 \cos cos\,\theta + m_2 v_2 \cos cos\,\theta \ldots (2)$

Also, K.E is conserved as the collision is elastic,

$\therefore \frac{1}{2} m_1 v^2 = \frac{1}{2} m_1 v_1^2 + \frac{1}{2} m_2 v_2^2 \ldots (3)$

$from\ 1\ \&\ 2,$

$m_1 v = 2 m_1 v_1 \cos cos\,\theta \Rightarrow v_1 = \frac{v}{2 \cos cos\,\theta} \ldots (4)$

And $v_2 = \frac{m_1 v_1}{m_2} \Rightarrow v_2 = \frac{m_1 v}{2 m_2 \cos cos\,\theta} \ldots (5)$

$\therefore from\ (iii),\ (iv) \& (v) \Rightarrow$

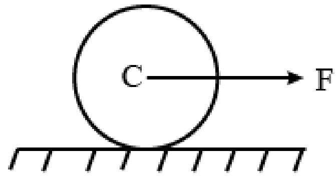
$m_1 v^2 = \frac{m_1 v^2}{4 cos^2\,\theta} + \frac{m_2 m_1^2 v^2}{4 m_2^2\, cos^2\theta}$

$\Rightarrow 1 = \frac{1}{4\, cos^2\theta}\left[1 + \frac{m_1}{m_2}\right] \Rightarrow \frac{m_1}{m_2} = 4 cos^2\,\theta - 1$

$\therefore \frac{m_1}{m_2} = 1 + 2\, cos\, 2\theta$

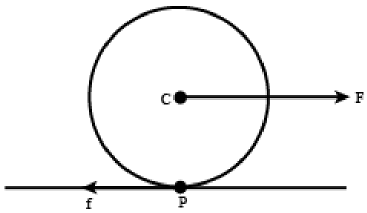
3. A solid cylinder of mass m is rolling without slipping on a rough horizontal surface, under the action of a horizontal force F such that the line of action of F passes through centre C of the cylinder. Choose the correct alternative.

A. Acceleration of centre of cylinder is $\frac{F}{m}$
B. Frictional force on cylinder acts forward
C. Magnitude of friction force is $\frac{F}{3}$
D. None of the above.



Answer (C)

Sol.



Let's calculate Torque about C,

$\tau = I\alpha$

Assuming radius of cylinder to be R,

$Rf = \left(\frac{1}{2}mR^2\right)\alpha$

$f = \frac{1}{2}mR\,\alpha. \quad (\because a = R\,\alpha)$

$\Rightarrow f = \frac{1}{2}ma...(1)$

In translatory motion,

$F_{net} = ma$

$F - f = ma$

From eqn. (1)

$F - \frac{1}{2}ma = ma$

$F = \frac{3}{2}ma$

$\Rightarrow a = \frac{2F}{3m}$

⇒ friction force, $f = \frac{1}{3}F$

⇒ friction acts backwards.

4. A motor pump is used to deliver water at a certain rate r from a given pipe. To obtain thrice as much water from the same pipe in the same time, the power of the motor has to be increased to

A. 3 times
B. 9 times
C. 27 times
D. 81 times

Answer (C)

Sol.

Mass flow rate, $\frac{dm}{dt} = \rho AV$

According to problem the new flow rate, $\left(\frac{dm}{dt}\right)' = 3\times\left(\frac{dm}{dt}\right)$

$A'v'\rho' = 3\times\rho Av$

$(A' = A\rightarrow Same\ pipe,\ \rho' = \rho\rightarrow Same\ liquid)$

$\Rightarrow v' = 3v$

Also,

Force $\Rightarrow \frac{F'}{F} = \frac{v'\left(\frac{dm}{dt}\right)'}{v\left(\frac{dm}{dt}\right)} = \frac{3v \times 3\left(\frac{dm}{dt}\right)}{v \times \frac{dm}{dt}}$

$\Rightarrow F' = 9F$

$\therefore Power \Rightarrow \frac{P'}{P} = \frac{F'v'}{Fv} = \frac{9F \times 3v}{Fv} = 27$

$\therefore P' = 27\,P$

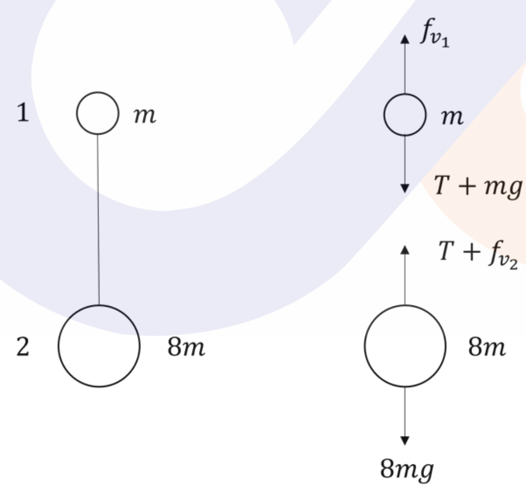
i.e., 27 times → option C.

5. Two small solid balls of masses m and 8m made up of same material are tried at the two ends of a thin weightless thread. They are dropped from a balloon in air. The tension T of thread during fall, after the motion of balls has reached steady state is

A. 2 mg
B. 3.5 mg
C. 4.5 mg
D. Zero

Answer (A)

Sol.



Viscos force, $F_v = 6\,\pi n\,r\,v\,...\,(1)$

For both solid balls, $\rho = \frac{8m}{\frac{4}{3}\pi r_2^3} = \frac{m}{\frac{4}{3}\pi r_1^3}$

$\therefore r_2 = 2r_1\,...\,(2)$

Motion of balls has reached steady state:

For equilibrium of the ball of mass 8m,

$T + F_{v_2} = 8\,mg\,...\,(3)$

For equilibrium of the ball of mass m,

$T + mg = f_{v_1}\,...\,(4)$

Also, from eqn. (1) & (2) $\Rightarrow F_{v_2} = 2\,F_{v_1}$

∴ from eqn (3) $\Rightarrow T + 2F_{v_1} = 8mg$

$\Rightarrow T + 2(T + mg) = 8mg$

$\Rightarrow 3T = 6mg$

$\Rightarrow T = 2mg$

→ Option (A).

6. Obtain the value of $\frac{e^2}{2\varepsilon_0 hc}$

A. 0.0073
B. 0.0073 m
C. 0.073 s^{-1}
D. 0.0346 m^{-1}

Answer (A)

Sol.

$$\frac{e^2}{2\epsilon_o hc} = \frac{(1.602\times10^{-19})^2}{2\times8.85\times10^{-12}\times6.62\times10^{-34}\times3\times10^8}$$

$= 7.3008\times10^{-3}$

$= 0.0073$

For dimensions:

$\epsilon_o = [M^{-1}L^{-3}T^4A^2]$

$h = [M^1L^2T^{-1}]$

$e = [A^1T^1]$

$c = [L^1T^{-1}]$

$$\therefore \frac{e^2}{\epsilon_o hc} = \frac{[A^2T^2]}{[M^{-1}L^{-3}T^4A^2][M^1L^2T^{-1}][L^1T^{-1}]}$$

$= [M^oL^0T^oA^o]$

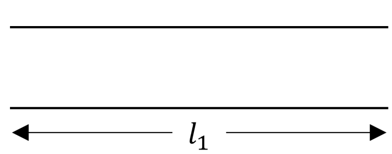
7. A sound source of fix frequency is in unison with an open end organ pipe of length 30.0 cm and a close end organ pipe of length 23.0 cm (both of same diameter). Both pipes are sounding their first overtone. If velocity of sound is 340ms^{-1}, frequency of sound source is nearly

A. 1000 Hz
B. 1062 Hz
C. 1100 Hz
D. 1018 Hz

Answer (B)

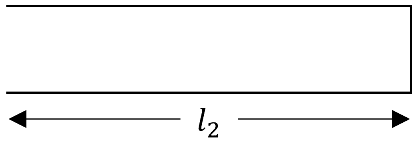
Sol.

For Open end organ pipe:



First overtone frequency, $f_1 = \frac{2v}{2l_1}$

For Closed organ pipe:



First overtone, $f_2 = \frac{3v}{4l_2}$

According to question, $\frac{2v}{2l_1} = \frac{3v}{4l_2} \Rightarrow \frac{l_1}{l_2} = \frac{4}{3} .\ .\ .\ .\ (1)$

Considering end correction, $\frac{l_1}{l_2} = \frac{30+(2\times(0.6\,r))}{23+0.6\,r}\ .\ .\ .\ (2)$

$\therefore from\ (1)\&\ (2) \Rightarrow r = \frac{2}{1.2}$

∴ end correction: 0.6 r = 0.6 × $\frac{2}{1.2} = 1\ cm$

$\therefore frequency,\ f_1 = \frac{2v}{2\times(l_1)} = \frac{340}{0.32} = 1062.5\ Hz$

$f_2 = \frac{3}{4}\left(\frac{v}{l_2}\right) = \frac{3}{4} \times \frac{340}{0.24} = 1062.5\ Hz$

8. Solar constant for Earth is 2.0 cal per cm^2 per minute. [1cal = 4.2 J]. Angular diameter of the Sun (as seen from the Earth) is $\frac{1}{2}^{\circ}$ (= half a degree). Treating Sun as a black body, its surface temperature is estimated to be nearly

A. 6000 K
B. 5800 K
C. 6200 K
D. 5500 K

Answer (A)

Sol:

Solar constant = $\frac{Energy}{Area\times time} = \frac{power}{surface\ area}$

∴ Solar constant, S = $\frac{\sigma\times 4\,\pi R_s^2\times T^4}{4\,\pi R^2}$

R_s = Radius of Sun

R = Mean orbital radius of Earth

$S = \frac{\sigma\times R_s^2\times T^4}{R^2}\ .\ .\ .\ .\ (1)$

$180^{\circ} \rightarrow \pi$

$1 \rightarrow \frac{\pi}{180}$

$\frac{1}{2} \rightarrow \frac{\pi}{180} \times \frac{1}{2}$

$\theta = \frac{\pi}{360}\ radian$

As θ is small, we can assume

$tan\,\theta \sim \theta$

$\Rightarrow \tan tan\,\frac{\theta}{2} = \frac{R_s}{R} \Rightarrow \frac{\theta}{2} = \frac{R_s}{R}\ ...\ (2)$

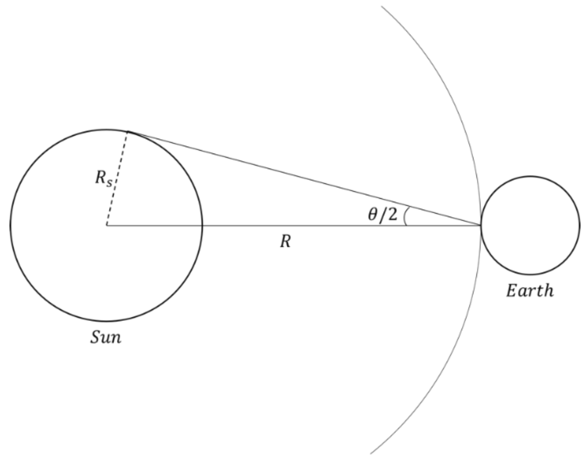
$\therefore from\ eqn.\ (1)\ \&\ (2),$

$S = \sigma \times \frac{\theta^2}{4} \times T^4$

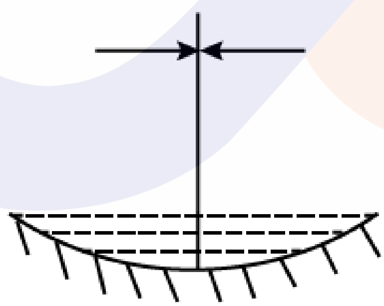
$\therefore T^4 = \left(\frac{4 \times S}{\sigma\theta^2}\right)$

$T^4 = \left(\frac{4 \times 2 \times 4.2 \times 10^4 \times 360^2}{60 \times 5.67 \times 10^{-8} \times \pi^2}\right)$

$T \cong 6000\ K$



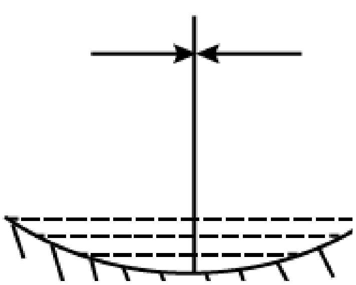
9. A concave mirror when placed in air has a focal length f = 20 cm. The mirror is now placed horizontally and filled with a thin layer of water having refractive index $\frac{4}{3}$. The object is placed horizontally near the principal axis at a distance d from the mirror such that a real, inverted image is formed at the same plane as the object, as shown in the figure. What is the value of d?



A. 30 cm
B. 20 cm
C. 15 cm
D. 40 cm

Answer (A)

Sol:



$\frac{1}{F_{eq}} = \frac{1}{F_m} - \frac{2}{F_l} \ldots . (1)$

Also, $\frac{1}{F_l} = \left(\frac{4}{3} - 1\right)\left(\frac{1}{\infty} - \frac{1}{-40}\right)$

$\therefore F_l = + 120\ cm$

$\therefore$ from eqn. (1) $\Rightarrow \frac{1}{F_{eq}} = \frac{-1}{20} - \frac{2}{120}$

$\therefore The\ combination\ acts\ as\ a\ mirror\ of\ focal\ length,\ F_{eq} = -15\ cm$

$\therefore d = 2F \Rightarrow d = 30\ cm$

$\rightarrow option\ (a)$

10. When a sample of atoms is irradiated by neutrons, radioactive atoms are produced at a constant rate R, which decay with decay constant λ. The number of radioactive atoms accumulated after an irradiation time t is given by

A. $N(t) = Rte^{-\lambda t}$
B. $N(t) = \frac{R}{\lambda}e^{-\lambda t}$
C. $N(t) = \frac{R}{\lambda}\left(1 - e^{-\lambda t}\right)$
D. $N(t) = Rt\left(1 - e^{-\lambda t}\right)$

Answer (C)

Sol.

$A \rightarrow B \rightarrow C$

Accumulation:

$dN = Rdt - \lambda N\ dt$

$\frac{dN}{dt} = R - \lambda N$

$\int_{O}^{N} \frac{dN}{R-\lambda N} = \int_{t=0}^{t} dt$

$\frac{\ln ln\ (R-\lambda N)}{-\lambda}\Big]_{o}^{N} = t$

$ln(R - \lambda N) - \ln ln\ R = -\lambda t$

$\ln ln\left(\frac{R-\lambda N}{R}\right) = -\lambda t$

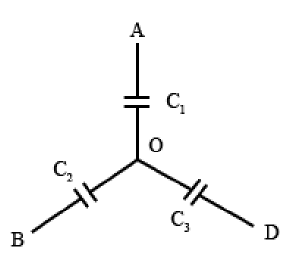
$\frac{R-\lambda N}{R} = e^{-\lambda t}$

$R - \lambda N = Re^{-\lambda t}$

$\lambda N = R\left(1 - e^{-\lambda t}\right)$

$N = \frac{R}{\lambda}\left(1 - e^{-\lambda t}\right)$

11. Three uncharged capacitors $C_1 = 2\mu F$, $C_2 = 3\mu F$ and $C_3 = 5\ \mu F$ are connected as shown in figure to one another at O and to points A, B and D at potentials $V_A = 300\ V$, $V_B = 200\ V$ and $V_D = 400\ V$ respectively the potential V_o at O is



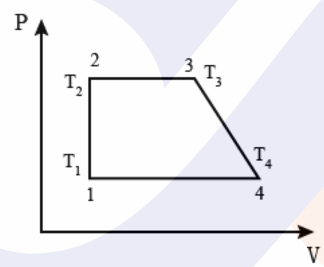
A. 300 V
B. 320 V
C. 240V
D. 280V

Answer (B)

Sol.

⇒ Using KCL
⇒ $Q_1 + Q_2 + Q_3 = 0$
⇒ $C_1(V_0 - 300) + C_2(V_0 - 200) + C_3(V_0 - 400) = 0$
⇒ $2(V_0 - 300) + 3(V_0 - 200) + 5(V_0 - 400) = 0$
⇒ $10V_0 = 3200$
⇒ $V_0 = 320\,Volt$

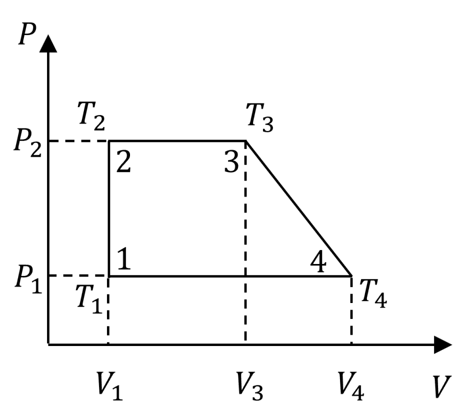
12. A cyclic process 1-2-3-4-1 consisting of two isobars 2 – 3 and 4 – 1, an isochor 1 – 2 and a process 3 – 4 represented by straight line on a P – V diagram, as shown in figure, involves n moles of an ideal gas. The gas temperatures at states 1, 2, 3 & 4 are T_1, T_2, T_3 and T_4 respectively. Also points 3 and 4 lie on the same isotherm. The work done by gas during the cycle is



A. $\frac{1}{2}nR(T_2 - T_1)\left(\frac{T_2}{T_1} + \frac{T_3}{T_4} - 2\right)$
B. $\frac{1}{2}nR(T_3 - T_2)\left(\frac{T_3}{T_2} + \frac{T_4}{T_1} - 2\right)$
C. $\frac{1}{2}nR(T_2 - T_1)\left(\frac{T_3}{T_1} + \frac{T_3}{T_2} - 2\right)$
D. Zero

Answer (C)

Sol.



Given, $T_3 = T_4$, $V_1 = V_2$, $P_1 = P_4$ and $P_2 = P_3$

$\Rightarrow$ Work done which is area of trapezium = $\frac{1}{2}\times\left[(V_4 - V_1) + (V_3 - V_2)\right](P_2 - P_1)$

$$= \frac{1}{2}\times\left[V_4 + V_3 - 2V_2\right]\times(P_2 - P_1)$$

$$= \frac{1}{2}\times\left[P_2V_2 - P_1V_2\right]\left[\frac{V_4}{V_2} + \frac{V_3}{V_2} - 2\right]$$

$$= \frac{1}{2}\times\left[P_2V_2 - P_1V_1\right]\left[\frac{V_4}{V_2} + \frac{V_3}{V_2} - 2\right]$$

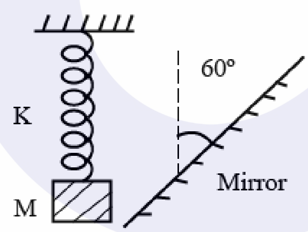
Using $PV = nRT$,

$$= \frac{1}{2}\times\left[nRT_2 - nRT_1\right]\left[\frac{V_4}{V_1} + \frac{V_3}{V_2} - 2\right]$$

$$= \frac{1}{2}\times nR\left[T_2 - T_1\right]\left[\frac{T_4}{T_1} + \frac{T_3}{T_2} - 2\right]$$

$$= \frac{1}{2}\times nR\left[T_2 - T_1\right]\left[\frac{T_3}{T_1} + \frac{T_3}{T_2} - 2\right]$$

13. An insect of negligible mass is sitting on a block of mass M, tied with a spring of force constant K. The block performs simple harmonic motion vertically with amplitude A in front of a mirror which is include at 60° with the vertical as shown. The maximum speed of insect relative to its image will be

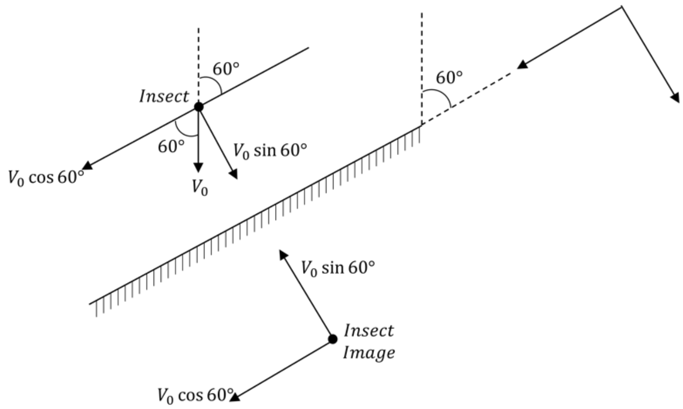


A. $2A\sqrt{\frac{K}{M}}$
B. $A\sqrt{\frac{3K}{M}}$
C. $A\sqrt{\frac{K}{M}}$
D. $zero$

Answer (B)

Sol.

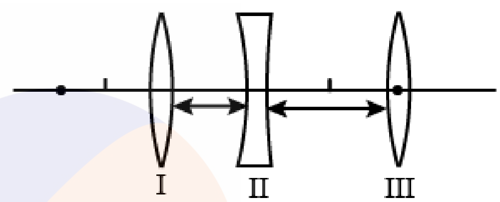
$$\Rightarrow V_{max}\ of\ insect = \omega A = \sqrt{\frac{K}{M}}A = V_0$$



$\Rightarrow at\ V_{max} \Rightarrow$

$\Rightarrow V_{rel} = V_0\sqrt{3} = \sqrt{3}\sqrt{\frac{K}{M}}$ A = $\sqrt{\frac{3K}{M}}A$

14. A concave lens of focal length 10 cm is placed between two convex lenses of focal length 10 cm and 20 cm at a separation of 5 cm between the first and second lens and 10 cm between the second and third lens. An object is placed at 30 cm in front of the first convex lens. The final image is formed beyond the third lens at a distance v from it. Then



A. v = 15 cm
B. v = ∞
C. v = 45 cm
D. v = 20 cm

Answer (D)

Sol.

For 1st lens ⇒ $\frac{1}{v_1} - \frac{1}{-30} = \frac{1}{10}$

$\Rightarrow v_1 = 15\ cm$

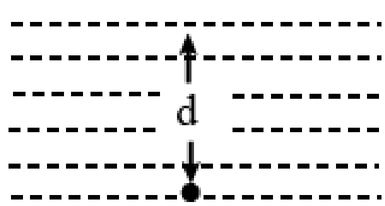
For 2nd lens ⇒ object distance = (15-5) = +10 cm (virtual object)

$\frac{1}{v_2} - \frac{1}{10} = \frac{1}{-10} \Rightarrow v_2 \rightarrow \infty \Rightarrow refracted\ rays\ are\ parallel$

For 3rd lens ⇒ Parallel rays converge at focus

⇒ v = 20 cm

15. A point source S of light is placed at a depth d below the surface of water in a large and deep lake. Fraction of light that escapes in space above directly from water (refractive index = μ) surface is given by



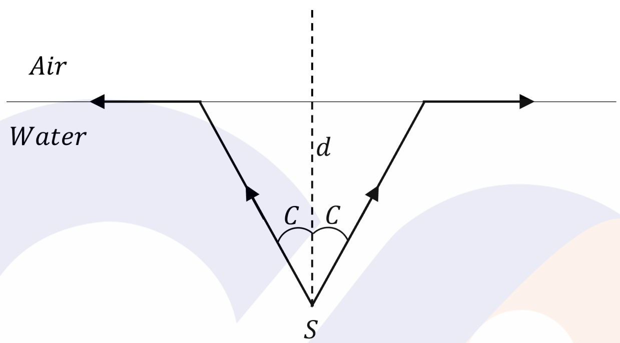
A. $\sqrt{1 - \frac{1}{\mu^2}}$
B. $\frac{1}{2}\sqrt{1 - \frac{1}{\mu^2}}$
C. $\frac{1}{2}\left\{1 - \sqrt{1 - \frac{1}{\mu^2}}\right\}$
D. Depends on d and increases with increasing d

Answer (C)

Sol.

→ Sin C = $\frac{1}{\mu}$ ⇒ $\cos \cos C = \sqrt{1 - \frac{1}{\mu^2}}$
→ Pint source emits light in all the directions.

⇒ But only the light within angle 'C' comes out.



For a cone with semi-vertical angle θ, its solid angle is, $2\pi\,(1 - \cos \cos \theta\,)$

⇒ So, fraction = $\frac{2\pi\,(1 - \cos\cos\ C)}{4\pi}$

⇒$\frac{1 - \cos\cos\ C}{2} = \frac{1}{2}\left[1 - \sqrt{1 - \frac{1}{\mu^2}}\right]$

16. A convex lens is held $45\ cm$ above the bottom of an empty tank. The image of a point object at bottom of tank is formed $36\ cm$ above the lens. Now a liquid is poured into the tank upto a height of $40\ cm$ above the bottom. It is found that distance of image of same point object at the bottom of the tank is $60\ cm$ above the lens. Refractive index of liquid is

A. 1.33
B. 1.37
C. 1.40
D. 1.60

Answer (D)

Sol.

Case 1 → without liquid in the tank, $v = 36\ cm$ & $u = 45\ cm$

Using Lens formula, $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$

$\Rightarrow \frac{1}{36} - \frac{1}{45} = \frac{1}{f}$

$\Rightarrow f = 20$

Case 2 → with liquid in the tank upto a height $40\ cm$

⇒ Apparent depth from surface of liquid $\Rightarrow d = \frac{40}{\mu}$

$\Rightarrow v = 60\ cm$ & object distance$(u) =- (5 + d)\ cm$

Using Lens formula, $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$

$\rightarrow \frac{1}{60} - \frac{1}{-(5+d)} = \frac{1}{20}$

$\Rightarrow d = 25\ cm$

$\Rightarrow d = \frac{40}{\mu} = 25 \Rightarrow \mu = 1.6$

17. A potential of $5\,V$ is applied across the face of a pure germanium plate of area $2\times10^{-4}m^{2}$ and of thickness $1.2\times10^{-3}m$. Concentration of carriers in germanium at room temperature is $1.6\times10^{6}m^{-3}$, Mobility of electrons and holes are $0.4\,m^{2}V^{-1}s^{-1}\ and\ 0.2\,m^{2}V^{-1}S^{-1}$ respectively. The current produced in germanium plate at room temperature,

A. $1.28\times10^{-13}A$
B. $1.28\times10^{-9}A$
C. $1.536\times10^{-13}A$
D. $6.4\times10^{-10}A$

Answer (A)

Sol.

In pure semiconductor ⇒ $n_e = n_h$

So ⇒ $n_e = n_h = 1.6\times10^{6}\ m^{-3}$

⇒ Current = $n_e Ae\left(V_{d_e}\right) + n_h Ae\left(V_{d_h}\right)$

$V_{d_e} =$ drift velocity of electrons

$V_{d_h} =$ drift velocity of holes

$\rightarrow Mobility = \mu = \frac{vd}{E},\ Electric\ field = \frac{\Delta V}{l}$

$\Rightarrow E = \frac{5}{1.2\times10^{3}}\ v/m$

$\rightarrow Current = (1.6\times10^{6})\ A\ e\left[V_{d_e} + V_{d_n}\right]$

$= \left(1.6\times10^{6}\right)\left(2\times 10^{-4}\right)\left(1.6\times10^{-19}\right)[m_1E + m_2E]$

$= 2.56\times2\times10^{-17} \times \frac{5}{1.2\times10^{-3}}\ [0.4 + 0.2]$

$= 1.28\times10^{-13}\ A$

18. Fission of one nucleus of $.^{235}U$ releases $200\,MeV$ energy in average. Minimum amount of $.^{235}U$ required to run

$1000\ MW$ reactor per year of continuous operation (assuming 30% efficiency) is

A. $1280\ ton$
B. $1.28\ ton$
C. $1.1\ ton$
D. $1.1\times10^{5} ton$

Answer (B)

Sol.

∵ Amount of energy required in a day

$= 1000\times24\times3600\ MJ$

→ no. of nuclie required per day = $\frac{24\times36\times10^{5}\ MJ}{200\times1.6\times10^{-19}\times(0.3)MJ}$

= 9×10^{24} $(at\ 30\%\ efficiency)$

→ no. of moles required per day $= \frac{Number\ of\ nuclei}{6\times10^{23}} = \frac{9\times10^{24}}{6\times10^{23}} = 15$

→ no. of moles required per year $= 15\times365 = 5475$

→ mass required = $(5475\times0.235)kg\cong1.28\ ton$

19. In a young's double slit experiment distance between slits is $d = 1\ mm$, Wavelength of light used is $600\ nm$ and distance of screen from the plane of slits is $D = 1\ m$. the minimum distance between two points on the screen where intensity falls to 75% of maximum intensity will be (Assume both sources of equal power).

A. 0.1 mm
B. 0.2 mm
C. 0.45 mm
D. 0.9 mm

Answer (B)

Sol.

$I_1 = \frac{75}{100} \times 4I_o = 3I_o = I_o + I_o + 2\,I_o \cos cos\,\Delta\,\phi$

$\Rightarrow \cos cos\,\Delta\,\phi = \frac{1}{2} \Rightarrow \Delta\,\phi = \frac{\pi}{3}$

$\Rightarrow \frac{2\,\pi}{\lambda}\,.\ \Delta x = \frac{\pi}{3} \Rightarrow \Delta x = \frac{\lambda}{6}$

$y = \frac{\Delta x D}{d} = \frac{\lambda\,D}{6d}$

Two such points will exist above and below the central maxima each at distance y from the central maxima,

→ min distance = $2y = \frac{\lambda D}{3d} = \frac{6\times10^{-7}\times1}{3\times10^{-3}}$

$= 0.2\ mm$

20. A ball is projected from horizontal ground. It attains a maximum height H on its projectile path and there after strikes a stationary smooth vertical wall and falls on ground vertically below the point of maximum height. Assume the collision with wall to be perfectly elastic, the height of the point on the wall where the ball strikes is

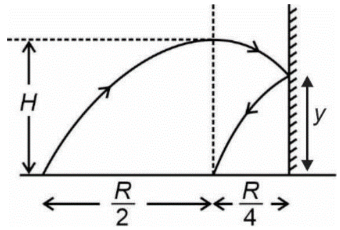
A. $3H/4$
B. $2H/3$
C. $H/2$
D. $4H/5$

Sol.

Answer (A)

⇒ Horizontal speed remains same throughout the motion, so the total distance travelled in horizontal

$\Rightarrow x = \frac{3R}{4}$, $\tan tan\, \theta = \frac{4H}{R}$

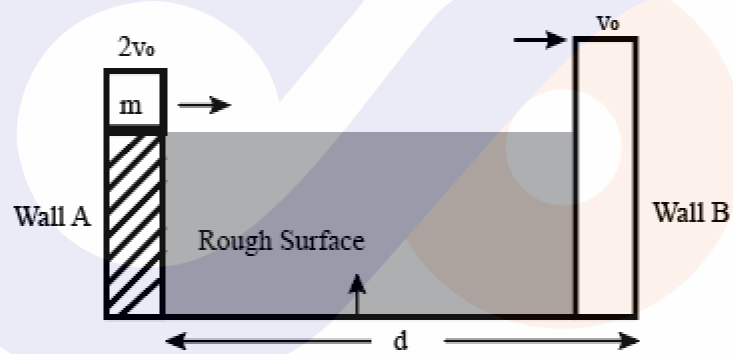


→ Trajectory ⇒ $y = x \tan tan\, \theta \left(1 - \frac{x}{R}\right)$

$\rightarrow y = \frac{3R}{4} \times \frac{4H}{R}\left(1 - \frac{3}{4}\right)$

$\rightarrow y = \frac{3H}{4}$

21. As shown in figure, a block of mass m is projected from wall A with velocity $2v_0$, on the rough surface with constant sliding friction to hit the wall B with velocity v_0. With what velocity same mass m should be projected to hit the wall B with same velocity v_0 if the surface is now moving upward with an acceleration of $a = 4g$



A. $2v_0$
B. $3v_0$
C. $4v_0$
D. $5v_0$

Answer (C)

Sol.

Case – 1

$v_0^2 = (2v_0) - 2a_1 d \Rightarrow a_1 = \frac{3v_0^2}{2d} = \mu g$

Here $a_1 = \frac{\mu N}{m} = \mu g\ \{N = mg\}$

Case – 2

Now $N = m(g + a) = 5mg \Rightarrow a_2 = \frac{\mu N}{m} = 5\mu g$

$a_2 = \frac{15v_0^2}{2d}$

Let u = initial velocity along the surface

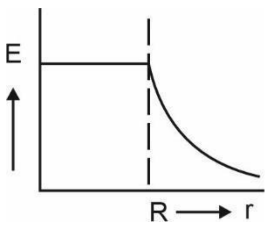
$v_0^2 = u^2 - 2a_2 d$

$v_0^2 = u^2 - 2(\frac{15v_0^2}{2d})d \Rightarrow v_0^2 = u^2 - 15v_0^2$

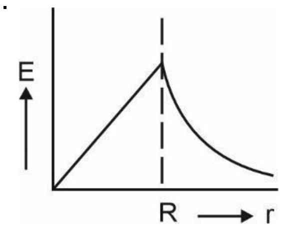
$\Rightarrow u^2 = 16v_0^2 \Rightarrow u = 4v_0$

22. A sphere of radius R, is charged with volume charge density ρ such that $\rho \propto r$ (r is distance from Centre). Variation of electric field E with r (For all values of r : $r \le R$ and $r > R$) is best represented by

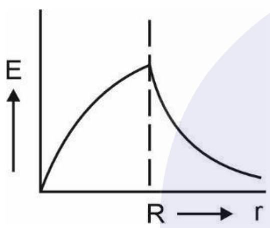
A.



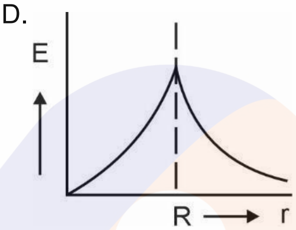
B.



C.

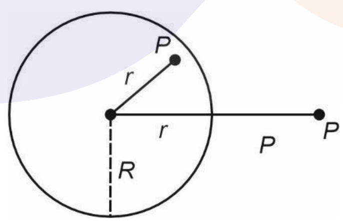


D.



Answer (D)

Sol. For a non-uniformly charged sphere the electric field $E(r)$ is given as



Using Gauss law, for $r < R$

$\oint \vec{E}.d\vec{A} = \frac{q_{en}}{\varepsilon_0} \Rightarrow E \times 4\pi r^2 = \frac{1}{\varepsilon_0}\int_0^r 4\pi x^2 kx.dx$

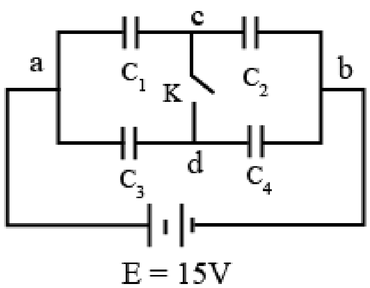
$E = \frac{kr^2}{4\varepsilon_0}$

For $r > R$

$E = \frac{1}{4\pi\varepsilon_0}\frac{q}{r^2}$

Therefore, the graph will be drawn as,

23. A system of capacitors $C_1 = 4\ \mu F,\ C_2 = 1\ \mu F,\ C_3 = 2\ \mu F$ and $C_4 = 3\ \mu F$ connected across a battery of emf E = 15 V is shown in figure. The charge that will flow, through the switch K, when it is closed, is

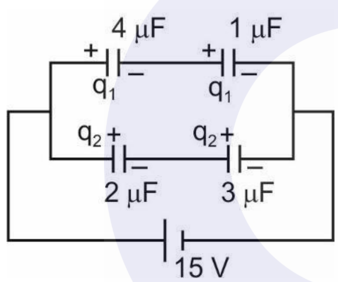


A. $15\ \mu C\ c\ to\ d$
B. $12\ \mu C\ c\ to\ d$
C. $6\ \mu C\ d\ to\ c$
D. $9\ \mu C\ d\ to\ c$

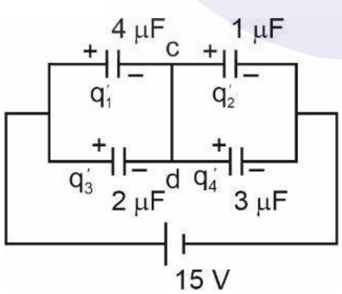
Answer (A)

Sol.

Before closing the switch



After switch is closed



Calculating from the diagram for the value of q_1, q_2, q_1', q_2' we get the values as

$q_1 = 15\frac{4}{5}\mu C,$

$q_2 = 15\frac{6}{5}\mu C,$

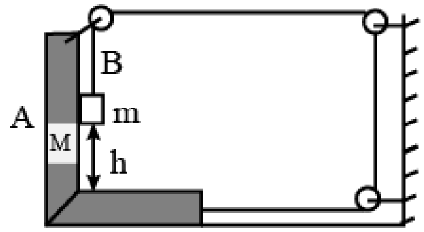
$q_1' = 24\ \mu C,$

$q_2' = 9\ \mu C$

Initially plates of capacitors which are connected to c has total charge 0 before switch is closed and net

charge of
$-24 + 9 = 15\ \mu C$ after the switch is closed thus total charge flown is $15\ \mu C$ from c to d.

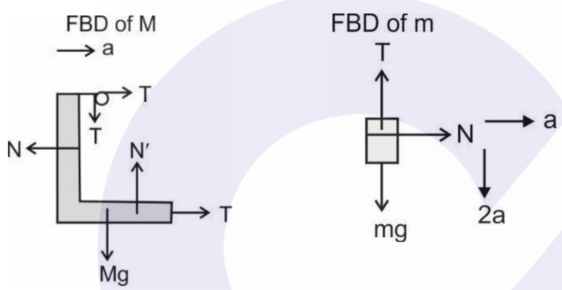
24. A simplification of a kind of interlock is shown in figure. All surfaces are smooth and frictionless. The body m has a mass m = 1 kg and the block M = 15 kg. The time 'm' takes to reach the base if it is released at height h = 4 meter above the base of M, is [use g = $10\ ms^{-2}$]



A. 1 s
B. $\sqrt{3}\ s$
C. 2 s
D. $2\sqrt{2}\ s$

Answer (C)

Sol.



From the FBDs we can write the equations as,

2T – N = Ma …(i)

N = ma …(ii)

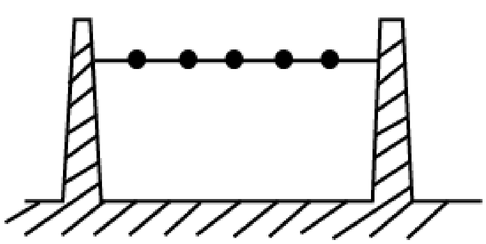
mg – T = 2ma …(iii)

On solving, 2mg = (M + 5m)a ⇒ a = 1 ms^{-2}

So, the acceleration of block in downward motion is 2a = 2 ms^{-2}, so time taken is

$$t = \sqrt{\frac{2 \times 4}{2}} = 2\ s$$

25. A number n of identical balls, each of mass m and radius r are stringed like beads at random and at rest along smooth, rigid horizontal rod of length L mounted between immovable supports; $\frac{r}{L}$ is small but not negligible.



Collision between balls, or between balls and supports, are perfectly elastic. One of the balls is struct horizontally so as to acquire a speed v. Resulting outward force felt by supports, averaged over a long time, is

A. $\frac{mv^2}{2(L-2nr)}$
B. $\frac{mv^2}{(L-2nr)}$
C. $\frac{2mv^2}{(L-2nr)}$
D. $\frac{mv^2}{L}$

Answer(B)

Sol. Total distance available for motion of beads = $L - 2nr$

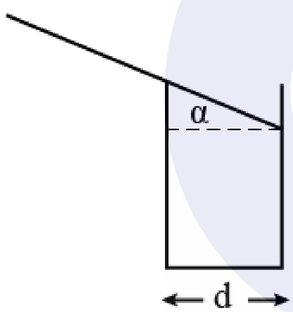
Change in momentum of a bead after collision = $2mv$

Avg time b/w two collision with one of the supports

$$t_{avg} = \frac{2\,(L-2nr)}{v}$$

$$F_{avg} = \frac{2mv}{\frac{2(L-2nr)}{v}} = \frac{mv^2}{L-2nr}$$

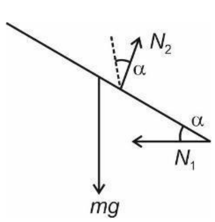
26. A cylindrical tumbler of diameter d has smooth sides and smooth edge. A thin of length L is balanced on the edge of the tumbler as shown in figure. The angle α that the rod makes with horizontal for this trick to work is



A. $sin^{-1}\left(\frac{d}{L}\right)^{\frac{1}{2}}$
B. $cos^{-1}\left(\frac{2d}{L}\right)^{\frac{1}{3}}$
C. $cos^{-1}\left(\frac{d}{L}\right)^{\frac{1}{3}}$
D. $sin^{-1}\left(\frac{2d}{L}\right)^{\frac{1}{2}}$

Answer (B)

Sol.



Writing the equation of NLM,

$N_2 \cos cos\, \alpha\ = mg$ ... (1)

Taking torque from point of application of N_1,

$N_2 d \sec sec\, \alpha\ = mg \frac{L}{2} \cos cos\, \alpha$ ... (2)

On solving, $\alpha\ = \frac{2d}{L}$

$$\alpha = cos^{-1}\left(\frac{2d}{L}\right)^{\frac{1}{3}}$$

27. End A of a uniform thin rod of length $2L$ is in boiling water (100 °C) and end B is in melting ice ($0\ °C$). P and Q are two points at distance $\frac{L}{2}$ from A and B respectively. A similar bent rod of length $\frac{3L}{2}$ of same material and equal cross section is joined to rod AB between points P and Q as shown in figure. Then

A. Temperature at P will increase and that at Q will decrease
B. Rate of flow of heat will increase by 25%
C. Rate of flow of heat will decrease by 20%
D. Rate of heat flow will increase by 37.5%

Answer (B)

Sol.

Let the thermal conductivity of material is *K* and cross-section area is *A.*

Before joining the bent rod

$T_P = 75°C, T_Q = 25°C$ and $\frac{dQ}{dt} = \frac{T_0 KA}{2L}$

After joining the bent rod net thermal resistance of combination is

$$R_{Th} = \frac{L}{2KA} + \frac{\frac{3L}{2KA} \times \frac{L}{KA}}{\frac{3L}{2KA} + \frac{L}{KA}} + \frac{L}{2KA} = \frac{L}{KA} + \frac{3L}{5KA} = \frac{8L}{5KA}$$

$$\frac{dQ'}{dt} = \frac{5T_0 KA}{8L}$$

So, percentage change in rate of heat flow

$$= \frac{\frac{5T_0 KA}{8L} - \frac{T_0 KA}{2L}}{\frac{T_0 KA}{2L}} \times 100 = 25\%$$

28. Two stars of masses M and m (M = 2 m) separated by a distance d = 3 astronomical unit, revolve in circular orbit about their centre of mass with a period of 2 years. If M_s is mass of Sun then

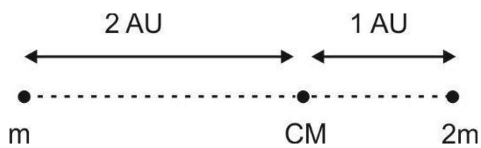
A. m = 2.25 M_s
B. m = 1.25 M_s
C. m = 2.50 M_s

D. m = 4.50 M_s

Answer (A)

Sol. Both the planets are revolving around each other in circular orbit, so

$$\frac{G2m^2}{3^2} = m\omega^2 \times 2$$



$$\omega = \sqrt{\frac{GM}{9}} \Rightarrow T = 2\,years = 2\pi\sqrt{\frac{9}{GM}} \Rightarrow \sqrt{GM} = 3\pi$$

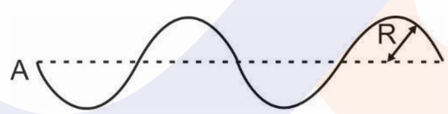
For Earth, $T' = 2\pi\sqrt{\frac{1}{GM_S}} = 1\,year$

So, $\sqrt{GM_S} = 2\pi$

$$\sqrt{Gm} = \frac{3}{2}\sqrt{GM_S}$$

So, $m = \frac{9}{4}M_S = 2.25M_S$

29. A thin uniform rod of mass M is bent into four adjacent semicircles of radius of curvature R lying in same plane. Moment of inertia of the bent rod about an axis through one end A and perpendicular to plane of the rod is



A. $\frac{17}{2}$ MR^2
B. 44 MR^2
C. 22 MR^2
D. $\frac{43}{2}$ MR^2

Answer (C)

Sol. If we complete all the semi-circular rings into a complete ring then moment of inertia about given axis will be

$$I = \left(\frac{M}{2}R^2 + \frac{M}{2}R^2\right) + \left(\frac{M}{2}R^2 + \frac{M}{2}(3R)^2\right) + \left(\frac{M}{2}R^2 + \frac{M}{2}(5R)^2\right) + \left(\frac{M}{2}R^2 + \frac{M}{2}(7R)^2\right) = 44MR^2$$

All the rings are half so desired moment of inertia is

$$I_0 = \frac{I}{2} = 22MR^2$$

30. Three point charges + q, −2q and +q are placed on x-axis at x = −d, x = 0 and x = +d respectively. The value of electric field at a point P on x axis at $x = r\ (r \gg d)$ is given by $E = \frac{1}{4\pi\varepsilon_0}\frac{aQ}{r^n}$ (Here $Q = 2qd^2$). Then

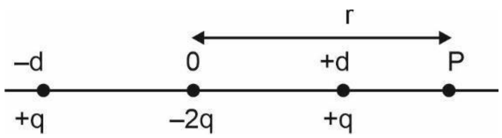
A. a = 3, n = 3
B. a = 6, n = 4

C. a = 3, n = 4

D. a = $\frac{3}{2}$, n = 4

Answer (C)

Sol.



$$E_{net} = \frac{1}{4\pi\varepsilon_0}\left(\frac{q}{(r+d)^2} - \frac{2q}{r^2} + \frac{q}{(r-d)^2}\right)$$

$$= \frac{q}{4\pi\varepsilon_0}\left(\frac{2(r^2+d^2)}{(r^2-d^2)} - \frac{2}{r^2}\right)$$

$$= \frac{2q}{4\pi\varepsilon_0}\left(\frac{r^4+r^2d^2-r^4+2r^2d^2-d^4}{r^2(r^2-d^2)^2}\right)$$

$$= \frac{2q}{4\pi\varepsilon_0}\left(\frac{3r^2d^2-d^4}{r^2(r^2-d^2)^2}\right)$$

$$= \frac{2qd^2}{4\pi\varepsilon_0 r^4} \quad (r \gg d) \qquad = \frac{3Q}{4\pi\varepsilon_0 r^4}$$

31. The frequency of the transverse oscillations of a proton (mass M) trapped in a cylindrical relativistic electron beam of circular cross section of radius R and current *I* is given by [assume that speed v of relativistic electrons ≈ c (the speed of light in vacuum) and ignore magnetic effect

A. $\frac{1}{2\pi R}\sqrt{\frac{eI}{2\pi\varepsilon_0 Mc}}$

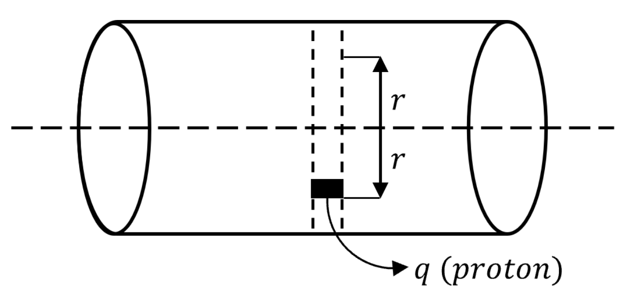
B. $\frac{1}{2\pi R}\sqrt{\frac{2\pi\varepsilon_0 I}{Mc}}$

C. $\frac{1}{R}\sqrt{\frac{2\pi\varepsilon_0 Mc}{eI}}$

D. $\frac{1}{2\pi\varepsilon_0}\sqrt{\frac{2\pi\varepsilon_0 Mc}{eI}}$

Answer (A)

Sol.



We know that,
Current through a conductor is $I = neAv_d$
Where,
n = Number of charges e per unit volume

e = Charge of an electron
A = Cross−sectional area
v_d = drift velocity

Equivalent charge density,
$\rho_0 = ne = \frac{I}{AC}$ (if $v_d = c$)

Now, electric field inside a solid uniformly charged long cylinder is
$E(r) = \frac{\rho_0}{2\varepsilon_0}.r = \frac{I}{2A\varepsilon_0 c}.r$
Now, force on proton,
$Ma = eE$

$a = \frac{eI}{2MA\varepsilon_0 c}.r$

Comparing with the standard equation of SHM, $|a| = \omega^2 r$

$\omega = \sqrt{\frac{eI}{2MA\varepsilon_0 c}}$

$f = \frac{\omega}{2\pi} = \frac{1}{2\pi}\sqrt{\frac{eI}{2MA\varepsilon_0 c}}$

$= \frac{1}{2\pi}\sqrt{\frac{eI}{2M\pi R^2\varepsilon_0 c}}$

$= \frac{1}{2\pi R}\sqrt{\frac{eI}{2\pi\varepsilon_0 Mc}}$

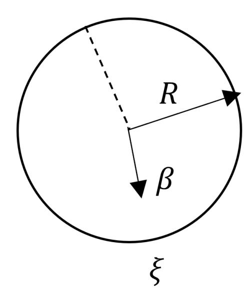
32. Current I flows through a long thin-walled metallic cylinder of radius R with a thin longitudinal slit of width ξ ($\xi \ll R$) running parallel to the axis of the cylinder. The magnetic induction B produced at any point on the axis of the cylinder is approximately

A. **B** = zero
B. **B** = $\frac{\mu_0 I}{2\pi R^2}$
C. **B** = $\frac{\mu_0 I\xi}{4\pi^2 R^2}$
D. **B** = $\frac{\mu_0 I\xi}{2\pi R^2}$

Answer (C)

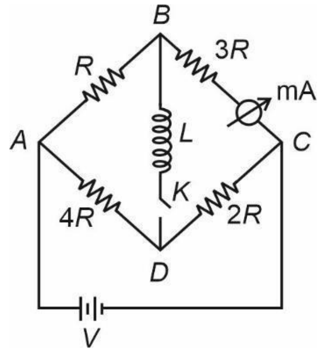
Sol.

Current for unit width = $\frac{i}{2\pi R}$
Field due to one element is multiplied by geometrically opposite element except the element opposite to slit,
Current through the opposite element is $di = \frac{i\,\xi}{2\pi R}$



$\therefore B = \frac{\mu_0\, di}{2\pi R} = \frac{\mu_0 I\,\xi}{4\pi^2 R^2}$

33. The reading of the ammeter, used in the electrical network shown below, is $20\ mA$, a long time after the key K is closed

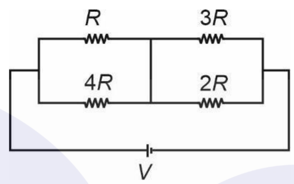


The reading of the same ammeter, immediately after the key was closed was

A. zero
B. $16\ mA$
C. $25\ mA$
D. $32\ mA$

Answer (C)

Sol. As $t \to \infty$ inductor is short $\Rightarrow$

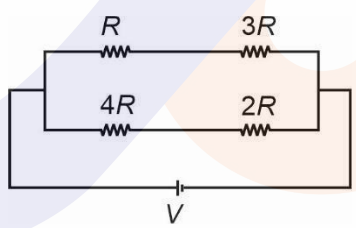


Since current in $3R$ is $20\ m$, $\Rightarrow V = \left[\frac{4R}{5} + \frac{6R}{5}\right] \times 50\ mA$

$= (100) mA.\ R$

As $t \to 0$ inductor is open $\Rightarrow$

$\therefore i = i_3 + i_A$



Reading of ammeter is given by $= \frac{V}{4R} = \frac{100R}{4R} = 25\ mA$

34. At the Earth's surface, a projectile is launched straight up at a speed of 10.0 km/s. Height to which it will rise is [g at surface of Earth $= 9.8\ ms^{-2}$ and radius of earth R = 6400 km]

A. 1.63×10^3 km
B. 1.56×10^4 km
C. 2.52×10^4 km
D. 5.1×10^3 km

Answer (C)

Sol.

Conserving energy:

$$-\frac{GMm}{R} + \frac{1}{2}mv^2 = -\frac{GMm}{R+h} + 0$$

$$\Rightarrow \frac{-GM}{R+h} = \frac{v^2}{2} - \frac{GM}{R}$$

$\Rightarrow -\frac{GM}{R^2}\frac{R^2}{R+h} = \frac{v^2}{2} - \frac{GM}{R^2}.R$

$\Rightarrow g\left[R - \frac{R^2}{R+h}\right] = \frac{v^2}{2}$

$\Rightarrow \frac{Rh}{R+h} = \frac{v^2}{2g} = \frac{(10^4)^2}{2\times9.8}$

$\Rightarrow \frac{h}{R+h} = \frac{10^8}{2\times9.8\times6.4\times10^6} = .797$

$\Rightarrow h = \frac{.797R}{0.2} = 2.52\times10^4\ km$

35. A small sphere of mass 2.00 g is released from rest in a large cylindrical vessel filled with oil. The resistive force due to viscosity of oil acting on sphere is proportional to its velocity. Sphere approaches a terminal speed of 5.00 cm/s. The time it takes the sphere to reach 90.0% of its terminal speed is approximately.

A. $3.22\ ms$

B. $5.10\ ms$

C. $10.2\ ms$

D. $11.7\ ms$

Answer (D)

Sol. Given f ∝ $v \Rightarrow f = kv,$ where k is constant and when $v = v_t,$ f =mg

$\Rightarrow kv_t = mg \Rightarrow k = \frac{mg}{v_t} = \frac{2\times10^{-3}\times10}{5\times10^{-2}}$

$k = 0.4\ Ns/m$

$\therefore f = \frac{2v}{5}$

a = $\frac{mg-f}{m} = g - \frac{f}{m} = g - \frac{2v}{5m}$

$\Rightarrow$ a = $\frac{dv}{dt} \Rightarrow \frac{dv}{dt} = \left(g - \frac{2v}{5m}\right)$

$\Rightarrow \int_0^t dt = \int_o^v \frac{dv}{g-\frac{2v}{5m}} \Rightarrow t = \frac{5m}{2} ln\left[g - \frac{2v}{5m}\right]_v^o$

$\Rightarrow t = \frac{5m}{2}\left[ln\ g - ln(g - \frac{2v}{5m})\right]$

$t = \frac{5m}{2}\left[\ln ln\left(\frac{5mg}{(5mg-2v)}\right)\right]$ ------------- (1)

For, $v = \frac{90}{100}v_t$

$v = 4.5\ cm/s$

Putting values in (1),

t = $5\times10^{-3}\left[ln\frac{5\times2\times10^{-2}}{(5\times2\times10^{-2})-(2\times4.5\times10^{-2})}\right]$

t = $5\times10^{-3} ln(10) = 2.303\times5\ ms$

t = 11.5 ms

36. A static point charge Q is located just above the centre C (δ 0) of a horizontal circle of radius R on its geometric axis, as shown in figure. The magnitude of electric flux through this circle is

A. Zero

B. $\frac{Q}{4\varepsilon_0}$

C. $\frac{Q}{2\varepsilon_0}$

D. $\frac{Q}{\varepsilon_0}$

Answer (C)

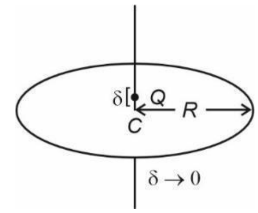
Sol.

Total flux due to Q = $\frac{Q}{\varepsilon_0}$

Flux below the end is almost same as above

$\therefore 2\phi = \frac{Q}{\epsilon 0} \Rightarrow \phi = \frac{Q}{2\epsilon 0}$

$\therefore$ flux through the given circle $= \frac{Q}{2\varepsilon_0}$



37. Three small identical neutral metal balls are at the vertices of an equilateral triangle. The balls are in turn touched to an isolated large charged conducting sphere whose centre is on a line perpendicular to the plane of triangle and passing through its centre. As a result the first and second balls have acquired charges $q_1$and q_2 respectively. The charge acquired by the third ball is [Assume that charge and potential of large spherical conductor change insignificantly in charging od the balls and that charges on balls are spherically symmetric]

A. $\frac{q_1^2}{q_2}$

B. $\frac{q_2^2}{q_1}$

C. $2q_2 - q_1$

D. $q_3 = q_2 = q_1$

Answer (B)

Sol. Let

r = radius of small ball

R =radius of large sphere

Q =Initial charge of large sphere

since on touching, potential would become same

$\Rightarrow \frac{q_1}{r} = \frac{Q-q_1}{R}$

$\Rightarrow q_1 = \frac{r}{R}Q$

$\Rightarrow q_2 = \frac{r}{R}\left[1 - \frac{r}{R}\right]Q$

$\& \; q_3 = \frac{r}{R}\left[1 - \frac{2r}{R}\right]Q \; [\because r \ll R]$

$\Rightarrow q_2^2 = \left[\frac{r}{R}Q\right]^2\left[1 - \frac{2r}{R}\right] = q_1 \cdot q_3$

$\Rightarrow q_3 = \frac{q_2^2}{q_1}$

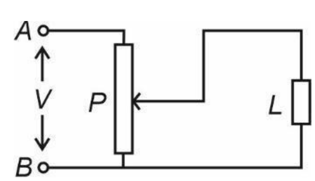
38. Voltage across the load L is controlled by using circuit as shown in figure. P is a potentiometer. Resistance R_L of the load and R_P of the potentiometer are equal to R. Load L is connected to the middle of potentiometer. Input voltage V is constant. If now R_L is doubled, the voltage across load will change by a factor

A. $\frac{5}{4}$

B. $\frac{7}{4}$

C. $\frac{8}{9}$

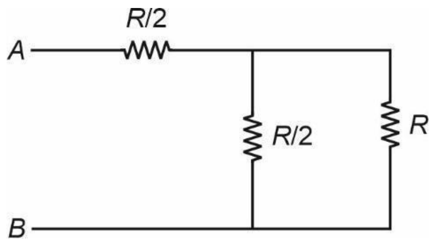
D. $\frac{10}{9}$



Answer (D)

Sol.

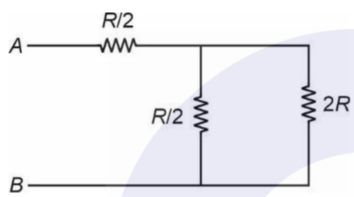
Initial:



Parallel combination $= \frac{R.\frac{R}{2}}{\frac{3R}{2}} = \frac{R}{3}$

$$\Rightarrow V_L = \frac{\frac{R}{3}}{\frac{R}{2}+\frac{R}{3}}V = \frac{2}{5}V$$

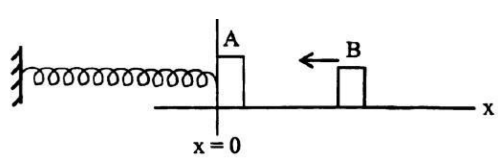
Final:



Parallel $= \frac{\frac{R}{2}.2R}{\frac{5R}{2}} = \frac{2R}{5}$

$$\Rightarrow V_L = \frac{\frac{2R}{5}}{\frac{2R}{5}+\frac{R}{2}}V = \frac{4V}{9} \Rightarrow \frac{4/9}{2/5} = 10/9$$

39. A small block A of mass 2 kg is attached to a spring of force constant 1200 Nm^{-1}, and rests on a smooth horizontal surface at x = 0 as shown in figure. A second block B of mass 1 kg slides along the surface towards A at 6 ms^{-1} and sticks to it. Assuming that the collision occurs at t = 0, position x (in meter) of block A as a function of time t is expressed as



A. x = 0.173 cos 20 t
B. x = 0.1 cos 40 π t
C. x = - 0.173 sin $\frac{\pi}{10}$ t
D. x = - 0.1 sin 20 t

Answer (D)

Sol. $\omega = \sqrt{\frac{k}{(mA+ms)}} = \sqrt{\frac{1200}{2+1}}$ = 20 rad/s

When B stick to A, $V_c = \frac{m_B/v_B}{m_A+m_B} = \frac{1\times6}{2+1} = 2\ m/s$

We have, $V_c = A\omega$, where A is amplitude.

$\Rightarrow 2 = A\times20 \Rightarrow A = 0.1\ m$

As the combined mass is merging initially towards –ve x –axis, from x =0

$x = -A \sin sin\ \omega t \Rightarrow x = -0.1 \sin sin\ 20\ t$

40. Two plane glass testing slides each of surface area A are stuck with each other by a small water drop squeezed between them as an extremely thin film of thickness d. If the surface tension of water be T and the contact be zero, then the force required to pull apart the two glass plates will be

A. $\frac{8TA}{d}$

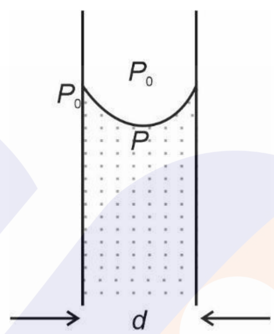
B. $\frac{4TA}{d}$

C. $\frac{2TA}{d}$

D. $\frac{TA}{d}$

Answer (C)

Sol.



$\Rightarrow P_0 - P = T\left\{\frac{1}{\frac{d}{2}}\right\} = \frac{2T}{d}$

$\Rightarrow$ Force needed $= \frac{2T}{d}\times area = \frac{2TA}{d}$

41. The rate of flow of a certain liquid of viscosity η through a horizontal capillary of length l and radius r is Q when the pressure head at the inlet is just twice the atmospheric pressure. The rate of flow of the same liquid through another capillary of length $2l$ and radius 2r when the inlet pressure head is 4 times the atmospheric pressure will be (The outlet being open to atmosphere in each case)

A. $24\ Q$

B. $16\ Q$

C. $8\ Q$

D. $4\ Q$

Answer (A)

Sol.

Rate of flow = $\frac{\pi \Delta P r^4}{8\eta l}$

$\Rightarrow\ Q = \frac{\pi P_0 r^4}{8\eta l}$ $[Let\ the\ atmospheric\ pressure\ be\ P_o]$

$Q' = \frac{\pi (3P_o)(2r)^4}{8\eta(2l)}$

$\Rightarrow Q' = 24\ Q$

42. A uniform rod of the material of Young's modulus Y is pushed over a smooth horizontal surface by a constant horizontal force F. The area of cross-section of the rod is A. The compressional strain in the rod is

A. $\frac{F}{AY}$

B. $\frac{F}{2AY}$

C. $\frac{3F}{2AY}$

D. $\frac{2F}{AY}$

Answer (B)

Sol.

When force is applied at one end, we know that extension

$\Delta l = \frac{Fl}{2AY}$

$\Rightarrow Average\ strain = \frac{\Delta l}{l} = \frac{F}{2AY}$

43. A total charge Q is uniformly distributed over a non – conducting ring of radius r. There is a time varying magnetic field perpendicular to its plane and changing at the uniform rate of $\frac{dB}{dt}$. The magnitude of induced tangential electric field E on ring is

A. $r\frac{dB}{dt}$

B. $r^2\frac{dB}{dt}$

C. $\frac{1}{2}r\frac{dB}{dt}$

D. $\frac{1}{2}r^2\frac{dB}{dt}$

Answer (C)

Sol.

$Faraday's\ Law\ of\ EMI$:

$\oint \vec{E}\vec{dl} = \left|-\frac{d\phi}{dt}\right|$

$\Rightarrow E.2\pi r = \pi r^2\frac{dB}{dt}$

$\Rightarrow E = \frac{r}{2}.\frac{dB}{dt}$

44. DC emf of 15 V is applied to a circuit containing 5 H inductance and $10\ \Omega$ resistance in series at t = 0. The ratio of the currents in the circuit at t = 0.5 sec and at t = 1.0 sec is

A. $\frac{e^2}{e^2-1}$

B. $\frac{\sqrt{e}}{\sqrt{e}-1}$

C. $\frac{e}{e+1}$

D. $\frac{1}{e}$

Answer (C)

Sol.

Time constant, $\frac{L}{R} = \frac{5}{10} = 0.5\ s$

i(t) = $\frac{E}{R}\left[1 - e^{-t/\tau}\right]$

$\Rightarrow \frac{i_{t=0.5\ s}}{i_{t=1\ s}} = \frac{1-e^{-1}}{1-e^{-2}} = \frac{1-\frac{1}{e}}{1-\frac{1}{e^2}} = \frac{(e-1)e}{e^2-1}$

$\Rightarrow \frac{e}{e+1}$

45. An insulating rod of length/ carries charge q distributed uniformly ll over its length. The rod is pivoted at its midpoint and is rotated at a frequency f (in Hz) about an axis perpendicular to the rod passing through the point at the pivot. The magnetic moment of the system is

A. $\frac{1}{12}\pi q f l^2$

B. $\frac{1}{6}\pi q f l^2$

C. $\frac{1}{3}\pi q f l^2$

D. $\pi q f l^2$

Answer (A)

Sol.

We know, $\frac{M}{L} = \frac{q}{2m}$

M = magnetic moment, L = Angular Momentum

q= charge, m = mass

$\Rightarrow \frac{M}{\frac{ml^2}{12}.2\pi f.} = \frac{q}{2m}$

$\Rightarrow M = \frac{ql^2 2\pi f}{2\times 12} = \frac{\pi f q l^2}{12}$

46. A circular loop of radius r is placed inside another circular loop of radius R (R >> r). The loops are coplanar and concentric. The manual inductance (M) of the system is proportional to

A. $\frac{r}{R}$

B. $\frac{r^2}{R}$

C. $\frac{R^2}{r}$

D. $\frac{r^2}{R^2}$

Answer (B)

Sol.

Let I be the Current in loop of radius R

Magnetic field at the centre of the loop of radius R, $B_C = \frac{\mu_0 I}{2R}$

Magnetic flux linked with the smaller loop, $\phi = BA = \frac{\mu_0 I}{2R} \times \pi r^2 = \frac{\mu_0 I \pi r^2}{2R}$

Magnetic flux linked with the smaller loop can also be written as, $\phi = MI$

$\Longrightarrow MI = \frac{\mu_0 I \pi r^2}{2R}$

$\Longrightarrow M = \frac{\mu_0 \pi r^2}{2R}$

$\Longrightarrow M \propto \frac{r^2}{R}$

47. The amplitude of the electric and magnetic fields associated with a beam of light of intensity 477.9 W/m^2 are, respectively,

A. $6\times10^{2}\frac{V}{m}and\ 2\times\ 10^{-6}\ T$

B. $3\times10^{2}\ \frac{V}{m}\ and\ 1\times\ 10^{-6}\ T$

C. $12\times10^{2}\frac{V}{m}\ and\ 4\times\ 10^{-6}\ T$

D. $6\times10^{2}\frac{V}{m}\ and\ 3\times\ 10^{-6}\ T$

Answer (A)

Sol.

$I\ =\frac{1}{2}\epsilon_0 E_0^2\, C$

$E_0\ =\sqrt{\frac{2I}{\varepsilon_0 c}}\ =\sqrt{\frac{2\times477.9}{8.85\times10^{-12}\times3\times10^{8}}}$

$\Rightarrow E_0\ =\ 6\times10^{2}\frac{V}{m}$

$B_0\ =\frac{E_0}{c}\ =\frac{6\times10^{2}}{3\times10^{8}}\ =\ 2\times10^{-6}\ T$

48. Given that the critical angle of incidence for total internal reflection within a transparent material when placed in air is 45°. The Brewster's angle of incidence for light propagating from air to the transparent material will be

A. 54.74° $B.\ 35.26^{\circ}$ $C.\ 25.26^{\circ}$ $D.\ 44.74^{\circ}$

Answer (A)

Sol.

Critical angle is given by, $\sin sin\ i_c\ =\frac{1}{\mu}$

$\Rightarrow\mu\ =\frac{1}{sin\ i_c}\ =\frac{1}{sin\ 45°}$

$\Rightarrow\ \mu\ =\sqrt{2}$

Brewster's angle is given by, $\theta\ =\ \mu$

$\Rightarrow\theta\ =\ tan^{-1}\sqrt{2}\ >\ 45°$

49. A hydrogen atom is in ground state $(n\ =\ 1)$. The magnetic field produced by revolving electron, at centre of atom is B_0. Atom is excited to state $_{n\ =\ 4}$. According to Bohr model, the correct alternative(s) is/are

A. Magnetic field at centre of atom for (n= 4) becomes $B_4\ =\frac{B_0}{64}$

B. Energy absorbed by atom in going from (n=1) to (n=4) is 12.75 eV

C. Change in magnitude of angular momentum of electron is $\frac{3h}{2\pi}$

D. Assume that this excited atom $(n\ =\ 4)$ is at rest and it makes transition to ground state $(n\ =\ 1)$ in a single quantum jump of an electron, (Take mass of atom $M_H\ =\ 1.67\times10^{-27}\ Kg)$ the recoil speed of atom will be nearly $v\ =\ 4.1\ ms^{-1}$.

Answer (B, C & D)

Sol.

$$B = \frac{\mu_0 ef}{2r} = \frac{\mu_0 e\, v}{2r.2\pi r}$$

$v \propto \frac{1}{n}$

$$r \propto n^2$$

$\Rightarrow B \propto \frac{1}{n^5}$

$\Rightarrow \frac{B(n=4)}{B_0} = \frac{1}{4^5}$

∴(A) is incorrect

\

Energy of n^{th} state of Hydrogen atom is given by,

$E_n = -\frac{13.6}{n^2}\, eV$

$$\Rightarrow \Delta E = 13.6\left[1 - \frac{1}{4^2}\right] eV = 13.6 \times \frac{15}{16} = 12.75\, eV$$

⇒(B) is correct

Angular momentum of an electron in the n^{th} state is given by,

$L = \frac{nh}{2\pi}$

$$\Rightarrow \Delta|L| = \frac{4h}{2\pi} - \frac{h}{2\pi} = \frac{3h}{2\pi}$$

∴(C) is correct

Using Conservation of linear momentum,

$|p_{atom}| = |p_{photon}|$

$\Rightarrow Mv = \frac{h}{\lambda}$

Rydberg's formula is given by,

$\frac{1}{\lambda} = R\left(\frac{1}{n_f^2} - \frac{1}{n_i^2}\right)$

$\Rightarrow Mv = Rh\left(\frac{1}{n_f^2} - \frac{1}{n_i^2}\right)$

$\Rightarrow Mv = Rh\left(1 - \frac{1}{4^2}\right) = \frac{15}{16} Rh$

$\Rightarrow v = \frac{15}{16} \times \frac{1.097\times10^7 \times 6.6\times10^{-34}}{1.67\times10^{-27}}$

$\Rightarrow v = 4.1\, ms^{-1}$

∴ (D) is correct

50. In an experimental set up to study the photoelectric effect a point source of light of power 3.2 mW is used. The source emits mono energetic photons of energy 5 eV and is located at a distance d = 0.8 m from centre of a stationary metallic sphere of work function W = 3.0 eV. The radius of the sphere is R=8 mm. Assume that the sphere is isolated and photo electrons are instantly swept away after emission. Also assume that the efficiency off photoelectric emission is one for every 10^6 photons. In the present set up

A. The de Broglie wavelength of fastest moving photoelectron is nearly $8.7\,Å$

B. It is observed that after some time emission of photoelectrons from the surface of metal sphere is stopped, the charge on sphere just when the electron emission stops is $64\pi\epsilon_0 \times 10^{-3} C$

C. Time after which photo electric emission stops is nearly 111 s

D. The light source emits 4×10^{15} photons per second.

Answer (A, B, C, D)

Sol.

According to Einstein's photoelectric equation,

$K_{max} = E - \phi$

$\Rightarrow K_{max} = 5\,eV - 3\,eV = 2\,eV = \frac{p_{max}^2}{2m}$

De Broglie wavelength is given by,

$\lambda_{db} = \frac{h}{p_{max}}$

$\lambda_{db} = \frac{6.63\times10^{-34}}{\sqrt{2\times9.1\times10^{-31}\times2\times1.6\times10^{-19}}}$

$\lambda_{db} = \frac{6.63\times10}{7.63}\,Å = 8.7\,Å$

(A) is correct

Photoelectric emission will stop when,

$eV_0 = 2\,eV \Rightarrow V_0 = 2\,V$ (V_0 : Potential of sphere)

The charge when potential of sphere is V_0 is,

$2 = \frac{1}{4\pi\epsilon_0}.\frac{Q}{r} \Rightarrow Q = 8\pi\,\epsilon_0\left(8\times10^{-3}\right) = 64\pi\epsilon_0 \times 10^{-3}C$

(B) is correct

Total photoelectrons emitted $= \frac{64\pi\epsilon_0\times10^{-3}C}{1.6\times10^{-19}}$

Let No. of photons incident in sphere is N

As the efficiency of photoelectric emission is 1 in 10^6 photons,

$\Rightarrow N = 10^6.\frac{64\pi\epsilon_0 10^{-3}}{1.6\times10^{-19}} = 40\pi\epsilon_0 \times 10^{22}$

Let n = No. of photons emitted be second by the source,

$P = nE$

$\Rightarrow 3.2\times10^{-3} = n\times5\times1.6\times10^{-19}$

$\Rightarrow n = \frac{2}{5}\times10^{16} = 4\times10^{15}$

Calculating the time for N photons to incident in sphere,

$\Rightarrow \frac{n}{4\pi d^2}\pi R^2.t = N \Rightarrow t = \frac{N\,4d^2}{\eta R^2} = \frac{40\pi\epsilon_0\times10^{22}\times4\times(0.8)^2}{4\times10^{15}\times\left(8\times10^{-3}\right)^2}$

$\Rightarrow t = 4\,\pi\epsilon_0 \times 10^8 \times 10^4 = \frac{10^{12}}{9\times10^9} = 111\,s$

∴ (C) and (D) are correct

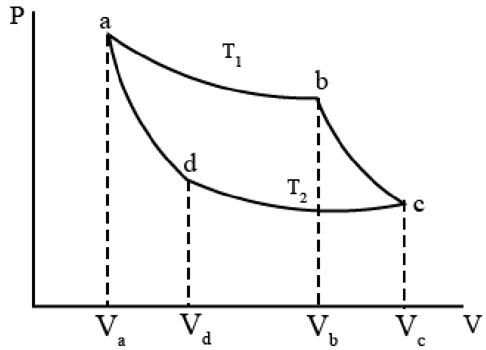
51. Two identical Carnot (cycles) engines operate between maximum and minimum temperatures T_1 and T_2 and volume limits, $V_a, V_b, V_c, \& V_d$ as shown in figure. Given that $\frac{V_c}{V_a} = e^3$ and $\frac{T_1}{T_2} = e$ (e is the base of natural logarithm). Engine 1 operates on mono atomic gas while engine 2 on diatomic gas. Choose correct alternatives

A. Ratio of volumes $\frac{V_{b,1}}{V_{b,2}} = e$

B. Ratio of work done per cycle for the two is $\frac{W_1}{W_2} = 3$

C. Ratio of work done per cycle for the two is $\frac{W_1}{W_2} = 1$

D. Ratio of efficiencies (η) of two engines $\frac{\eta_1}{\eta_2} = 1$



Answer (A, B, D)

Sol.

Efficiency in both cases remains same as efficiency depends on temperature only.
∴Option D is correct.

For adiabatic expansion,

$T_1 V_b^{\gamma-1} = T_2 V_c^{\gamma-1}$

For Engine 1,

$\Rightarrow eT_2 (V_{b1})^{\frac{2}{3}} = T_2 (e^3 V_a)^{\frac{2}{3}}$

$\Rightarrow V_{b1} = e^{\frac{3}{2}} V_a$

Similarly, for Engine 2,

$V_{b2} = e^{\frac{1}{2}} V_a$

$\Rightarrow \frac{V_{b1}}{V_{b2}} = e$

∴ Option A is correct.

Work done by the engine per cycle is given by,

$W = \eta Q_{abs}$

Heat is absorbed by the engine during isothermal expansion,

$\Rightarrow Q_{abs} = nRT_1 \ln ln \frac{V_b}{V_a}$

$\Rightarrow W = \eta nRT_1 \ln ln \frac{V_b}{V_a}$

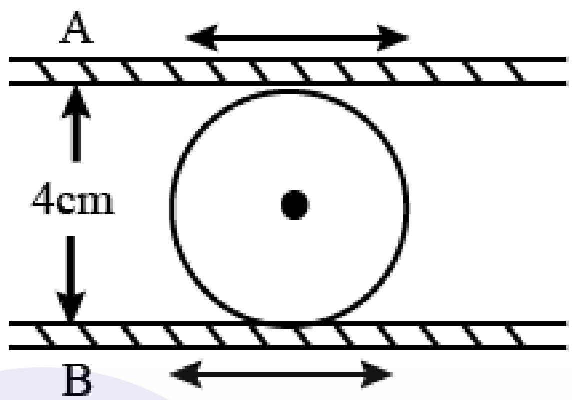
$\frac{W_1}{W_2} = \frac{\ln ln \frac{V_{b1}}{V_a}}{\ln ln \frac{V_{b2}}{V_a}} = \frac{\ln ln\, e^{\frac{3}{2}}}{\ln ln\, e^{\frac{1}{2}}} = 3$

∴ Option B is correct.

52. In a certain machine two steel plates are separated by a hardened steel cylindrical roller (see fig). In operation, the plates move back and forth horizontally, perpendicular to the axis of roller, and the roller rolls freely between plates without slipping on either one. At a particular instant plate A is moving with a speed of 18 cm sec^{-1} to the right and an acceleration of 30 cm sec^{-2} to the left, and the plate B is moving with a speed of 6 cm sec^{-1} to the

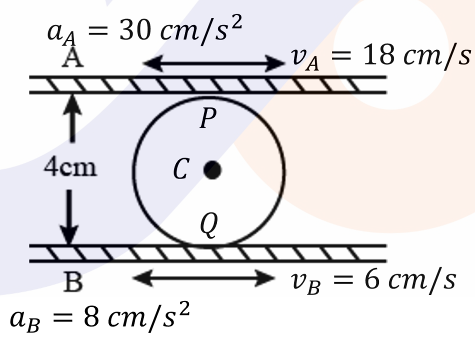
right and an acceleration of 8 cm sec^{-2} to the left. At that instant, for the roller

a) Its angular speed is 3 rad sec^{-1} clockwise
b) Its angular acceleration is 6 rad sec^{-2} clockwise
c) The linear speed of its axis is 12 cm sec^{-1} towards right
d) The linear acceleration of its axis is 20 cm sec^{-2} towards left



Answer (A,C)

Sol.



$\vec{V}_p = 18\hat{i}$

$\vec{V}_Q = 6\hat{i}$

Let the velocity of C is $V_c\hat{i}$ and angular velocity about C is $is\ \omega(-\hat{k})$

From the condition of pure rolling,

$v_p = v_c + \omega R$

$v_Q = v_c - \omega R$

$R = 2\ cm\ (given)$

$V_c + 2\omega = 18$

$v_c - 2\omega = 6$

$\Rightarrow v_c = 12\frac{cm}{s};\ \omega = 3\ rad/s$ (clockwise)

Let the acceleration of c is $a_c\ \hat{i}$ and angular acceleration about c is $\propto(-\hat{k})$

$\vec{a}_{p_t}$ (tangential acceleration of p)$=\vec{a}_A$

$\vec{a}_{Q_t}$ (tangential acceleration of Q)$=\vec{a}_B$

$a_{p_t} = a_c + R\alpha$

$a_{Q_t} = a_c - R\alpha$

$a_c + 2\propto \; =- \; 30$

$a_c - 2\propto \; =- \; 8$

$\Rightarrow a_c = 19(-\hat{i})cm/s^2$

$\propto \; = \frac{11}{2}(\hat{k})\text{cm}/s^2$

Correct option: (A), (C)

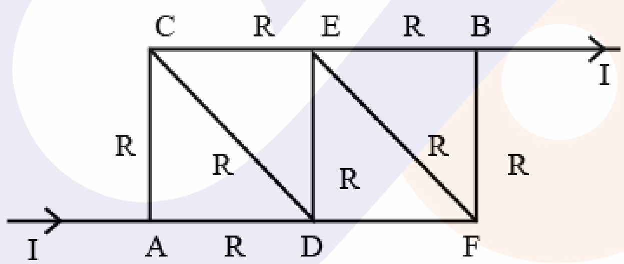
53. Each of 9 sides of frame A C D E F B has resistance R (Nine in all) A current I enters at A and leaves at B. Choose the correct alternatives.

A. Currents in branches CD and EF are zero.

B. Currents in branches CE and DF are each equal to $\frac{4}{11}I$

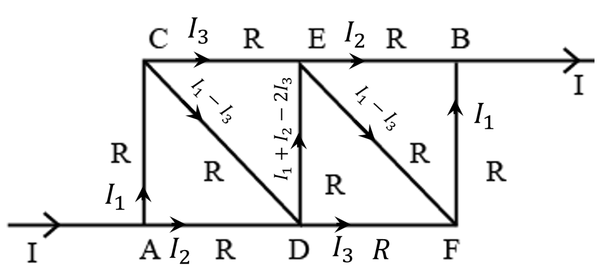
C. Effective resistance between A and B is $\frac{15}{11}\ R$

D. Effective resistance between A and B is $\frac{3}{4}\ R$



Answer (B,C)

Sol.



Based on the Input Output symmetry, the current distribution is shown in the figure.

$$V_A - V_B = V_A - V_c + V_c - V_E + V_E - V_B$$

$$V = I_1R + I_3R + I_2R$$

$$\Rightarrow I_1 + I_2 + I_3 = \frac{V}{R} --- (1)$$

$V_A - V_B = V_A - V_D + V_D - V_E + V_E - V_B$

$V = I_2R + (I_1 + I_2 - 2I_3)R + I_2R$

$I_1 + 3I_2 - 2I_3 = \frac{V}{R}$ --- (2)

$V_A - V_B = V_A - V_C + V_C - V_D + V_D - V_F + V_F + V_F - V_B$

$V = I_1R + (I_1 - I_3)R + I_3R + I_1R$

V = $3I_1R$

$\Rightarrow I_1 = \frac{V}{3R}$ --- (3)

From (1), (2) and (3)

$I_1 = \frac{V}{3R}$

$I_2 = \frac{2V}{5R}$

$I_3 = \frac{4V}{15R}$

$R_{eq} = \frac{V}{I} = \frac{V}{I_1+I_2} = \frac{15R}{11}$

Current through CD and EF = $I_1 - I_3 = \frac{V}{3R} - \frac{4V}{15R} = \frac{V}{15R}$

Current through CE and DF = $I_3 = \frac{4V}{15R}$

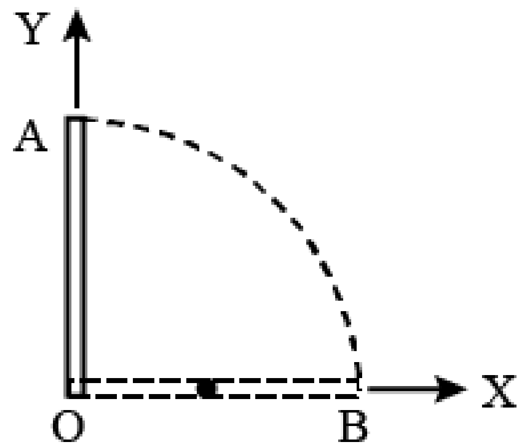
$I = I_1 + I_2 = \frac{11V}{15R}$

$I_3 = \frac{4I}{11}$

Correct option (B), (C)

54. A long uniform rod of length L and mass M is pivoted vertically on a horizontal, friction less pivot at its lower end. The rod is released from rest in its vertical position OA (see figure). It falls off without slipping at O. At the instant the rod is horizontal,

A. Its angular speed is $\sqrt{\frac{3g}{L}}$
B. Magnitude of its angular acceleration is $\frac{3g}{2L}$
C. Acceleration of its centre of mass $\vec{a}_{cm} = -\frac{3g}{4}\hat{j}$($\hat{j}$ unit vector in Y direction)
D. Reaction force at pivot $\frac{Mg}{4}\hat{j}$ (Take X, Y axis as shown)



Answer (A,B)

Sol.

According to conservation of total mechanical energy at vertical and horizontal position

$$Mgh = \frac{1}{2}I\omega^2$$

$$\frac{MgL}{2} = \frac{1}{2}\frac{ML^2}{3}\omega^2$$

$$\Rightarrow \omega^2 = \frac{3g}{L}$$

$$\Rightarrow \omega = \sqrt{\frac{3g}{L}}$$

∴ Option A is correct.

Torque about 0 at horizontal position

$$mg\frac{L}{2}\left(-\hat{k}\right) = I\vec{\propto}$$

$$Mg\frac{L}{2}\left(-\hat{k}\right) = \frac{ML^2}{3}\vec{\alpha}$$

$$\Rightarrow |\alpha| = \frac{3g}{2L}$$

∴ Option B is correct.

Centre of mass has two acceleration components. One in downward direction and other towards centre.

$$a_y = \propto\frac{L}{2} = \frac{3g}{4}$$

$$a_x = \omega^2\frac{L}{2}\left(-\hat{i}\right)(centripetal\ acceleartion)$$

∴ Option C is incorrect.

Taking two component of reaction force at point

Mg – $F_y = Ma_y$

$$F_y = Mg - Ma_y$$

$$F_y = mg - \frac{3mg}{4} = \frac{mg}{4}$$

$$\vec{F}_y = \frac{mg}{4}\hat{j}$$

$$F_x = M\omega^2\frac{L}{2}$$

$$F_x = M\frac{3g}{L} \times \frac{L}{2} = \frac{3Mg}{2}$$

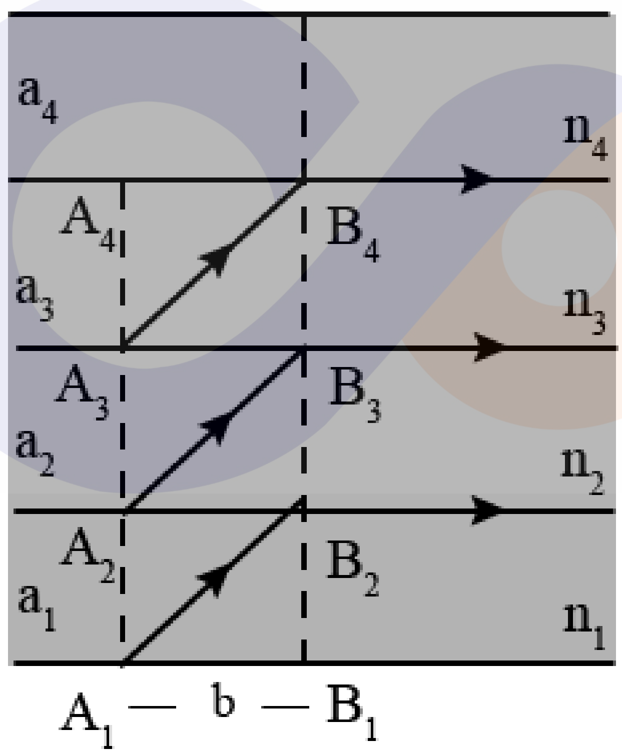
$F_x = \frac{3Mg}{2}(-\hat{i})$

$\therefore$ Option D is incorrect.

Correct option (A, B)

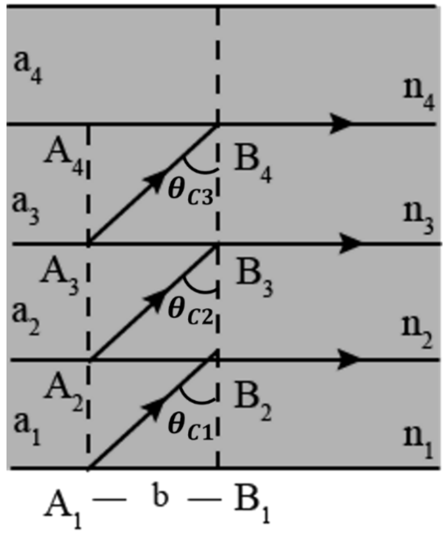
55. There are four layers of glass plates, placed on top of each other such that bottom one has thickness a_1 and refractive index $n_1 = 2.7$. Next one has thickness a_2 and refractive index $n_2 = 2.43$. The third one and the top one has thickness a_3 *and* a_4 and refractive indices n_3 *and* n_4 respectively. Three rays starting at the same moment from A_1, A_2 *and* A_3 reach pints B_2, B_3, B_4 at the same time, with their angles of incidence being critical angle. You are given $A_1B_1 = A_2B_2 = A_3B_3 = A_4B_4 = b = 10\ mm$. Choose correct statement (s).

A. $n_3 = 1.968$ B. $n_4 = 1.291$ C. $a_2 = 7.243\ mm$ D. $a_3 = 11.51\ mm$



Answer (A, B, C, D)

Sol.



In first interface between n_1 and n_2,

$$\sin sin\ \theta_{c1} = \frac{n_2}{n_1} = \frac{2.43}{2.7} = \frac{9}{10}$$

$$\Rightarrow \tan tan\ \theta_{c1} = \frac{9}{\sqrt{19}} = \frac{b}{a_1}$$

$$\Rightarrow a_1 = \frac{10\sqrt{19}}{9}\ mm$$

Time taken for the light from $A_1\ to\ B_2$ is given by,

$$t = \frac{A_1B_2}{v_1} = \frac{\sqrt{a_1^2+b^2}}{\frac{c}{n_1}}$$

Time taken for the light from $A_2\ to\ B_3$ is given by,

$$t = \frac{A_2B_3}{v_2} = \frac{\sqrt{a_2^2+b^2}}{\frac{c}{n_2}}$$

Given that, the time taken is same in all the three cases,

$$\Rightarrow \frac{\sqrt{a_1^2+b^2}}{\frac{c}{n_1}} = \frac{\sqrt{a_2^2+b^2}}{\frac{c}{n_2}}$$

$$\Rightarrow a_2 = 7.243\ mm$$

In second interface between n_2 and n_3,

$$\sin sin\ \theta_{c2} = \frac{n_3}{n_2}$$

Also, $\tan tan\ \theta_{c2} = \frac{b}{a_2}$

$$\Rightarrow n_3 = 1.968$$

Similarly, considering the time taken from $A_2\ to\ B_3$ and $A_3 to\ B_4$

$$\Rightarrow \frac{\sqrt{a_2^2+b^2}}{\frac{c}{n_2}} = \frac{\sqrt{a_3^2+b^2}}{\frac{c}{n_3}}$$

$\Rightarrow a_2 = 11.51\ mm$

In third interface between n_3 and n_4,

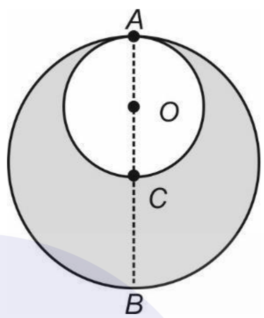
$\sin sin\ \theta_{c3} = \frac{n_4}{n_3}$

Also, $\tan tan\ \theta_{c3} = \frac{b}{a_3}$

$\Rightarrow n_4 = 1.291$

Correct option (A, B, C, D)

56. In an isolated asteroid of radius *R* and uniform density ρ, a spherical cavity of diameter *AC* = *R* is excavated,



where *C* is centre of asteroid. Choose correct alternative(s)

A. A ball just dropped from *A* will strike *C* with speed $v = 2R\sqrt{\frac{□□G}{3}}$

B. A ball dropped from *A* will reach *C* after time $t = \sqrt{\frac{3}{□□G}}$

C. Acceleration of ball dropped from *A* varies as its distance from *O* (centre of cavity)

D. Weight of a body placed at B (diametrically opposite to A) on the surface of the asteroid decreases by a factor of $\frac{7}{8}$ due to excavation of cavity.

Answer (A, B)

Sol.

Gravitational field inside the gravity is constant and directed parallel to $\vec{OC}$

$\vec{E} = \frac{2}{3}\pi G\rho R\left(-\hat{j}\right)$

So, particle move along AC with constant acceleration

$a = \frac{2}{3}\pi G\rho R$

Using equations of motion for constant acceleration,

$v_c^2 = 2\,a\,(AC)$

$v_c^2 = 2\times\frac{2}{3}\pi G\rho R\times R$

$v_c = 2R\sqrt{\frac{\pi\rho G}{3}}$

∴ option A is correct.

$AC = \frac{1}{2}at^2$

$R = \frac{1}{2} a t^2$

$\Rightarrow t = \sqrt{\frac{2R}{a}} = \sqrt{\frac{2R}{\frac{2}{3}\pi G\rho R}}$

$\Rightarrow t = \sqrt{\frac{3}{\pi\rho G}}$

∴ option B is correct.

As acceleration is constant, option C is incorrect.

$E_B = \frac{GM}{R^2}$ $(Without\ excavation)$

$E_B' = \frac{GM}{R^2} - \frac{G\left(\frac{M}{8}\right)}{\left(\frac{3R}{2}\right)^2} = \frac{GM}{R^2} - \frac{4}{9}\frac{GM}{R^2}.\frac{1}{8}$

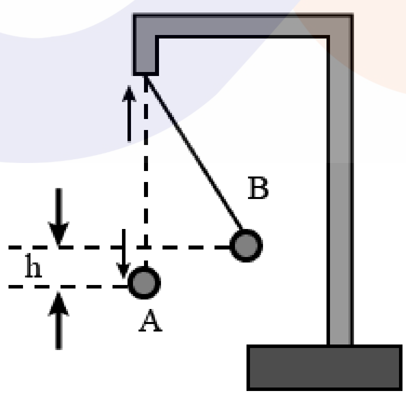
$E_B' = \frac{GM}{R^2} - \frac{GM}{18R^2} = \frac{17}{18}\frac{GM}{R^2}$

Weight decreases by factor $\frac{17}{18}$.

∴ option D is incorrect.

57. A small positively charged ball of mass m is suspended by a long insulating thread of negligible mass. Other positively charged small ball is oved very slowly from a large distance (along horizontal direction) until it is at original position A of first ball. As a result, the first ball rises by h to position B such that $h \ll l$. Choose the correct statement(s).

A. Electrostatic energy of system of charges is $2mgh$.
B. Total work done on system to bring two balls in their final position is mgh.
C. Total work done on the system to bring two balls in their final position is $3mgh$.
D. Work done on system to bring two balls in their final position does not depend on the magnitude of charges explicitly.



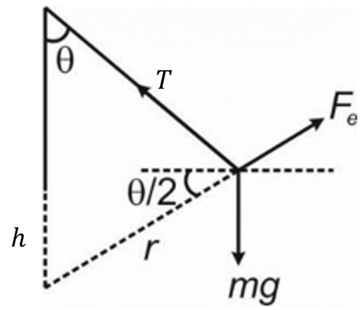
Answer (A, C, D)

Sol.

$W_{electrostatics} + W_{gravity} + W_{ex} = \Delta KE$

$W_e - mgh + W_{ex} = 0$

$W_e + W_{ex} = mgh$



Resolving forces as vertical and horizontal components at B,

$T \sin sin\, \theta\ = F_e cos\frac{\theta}{2} \Rightarrow F_e\ = \ 2T \sin sin\ \frac{\theta}{2}$

$T \cos cos\ \theta\ + F_e\ sin\ \frac{\theta}{2}\ \ = Mg$

As θ is small, $\cos cos\ \theta \approx 1\ \ and \sin sin\ \frac{\theta}{2}\ \ is\ neglected,$

$\Rightarrow T\ = \ mg$

From the figure, $\sin sin\ \frac{\theta}{2} = \frac{h}{r}$

$\Rightarrow F_e\ = \frac{2mgh}{r} = \frac{kq_1q_2}{r^2}$

$\Rightarrow U_e\ = \frac{kq_1q_2}{r} = \ 2mgh$

$\Rightarrow W_e\ =-\ U_e\ =-\ 2mgh$

$\Rightarrow W_{ext}\ = \ 3mgh.$

∴ Option A, C and D are correct.

58. A rope of mass m and length L is suspended vertically. A mass M is suspended from bottom of the rope. A transverse wave is produced on the rope, which travels the length of rope in time t choose the correct statement(s)

A. $t\ = \ 2\sqrt{\frac{L}{mg}}\,(\sqrt{M\ +\ m} - \sqrt{m})$

B. For m < < M, the time $t\ = \sqrt{\frac{mL}{Mg}}$

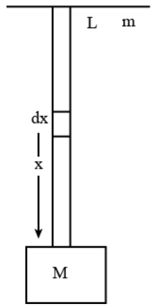
C. For M = 0 (i.e., no mass hanging), the time, $t\ = \sqrt{\frac{L}{g}}$

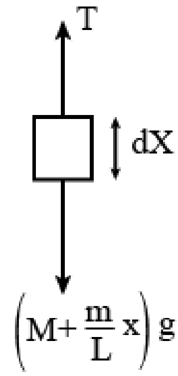
D. Time to travel the lower half of the rope by the wave is less than that to travel the upper half.

Answer (A, B)

Sol.

Considering a small element dx at a distance x from M,





$T = \left(M + \frac{m}{L}x\right)g$

Velocity of the wave in the string is given by,

$v = \sqrt{\frac{T}{\mu}} = \sqrt{\frac{\left(M+\frac{m}{L}x\right)g}{\frac{m}{L}}}$

$v = \sqrt{\left(\frac{ML}{m} + x\right)g}$

$\frac{dx}{dt} = \sqrt{\left(\frac{ML}{m} + x\right)g}$

$\int_0^L \frac{dx}{\sqrt{\left(\frac{ML}{m}+x\right)}} = \sqrt{g}\int_0^t dt$

$\left[2\sqrt{\frac{ML}{m} + x}\right]_0^L = \sqrt{gt}$

$\left[2\sqrt{\frac{ML}{m} + L} - 2\sqrt{\frac{ML}{m}}\right] = \sqrt{g}t$

$\Rightarrow t = 2\sqrt{\frac{L}{mg}}\left[\sqrt{M + m} - \sqrt{M}\right]$

∴ Option A is correct.

$For\ m \ll M,$

$t = 2\sqrt{\frac{L}{mg}}\sqrt{M}\left[\left(1 + \frac{m}{M}\right)^{\frac{1}{2}} - 1\right]$

$t = 2\sqrt{\frac{L}{mg}}\sqrt{M}\left[1 + \frac{m}{2M} - 1\right]$

$\Rightarrow t = \sqrt{\frac{mL}{Mg}}$

∴ Option B is correct.

$For\ M = 0,$

$t = 2\sqrt{\frac{L}{mg}}\left(\sqrt{M + m} - \sqrt{M}\right)$

$t = 2\sqrt{\frac{L}{g}}$

∴ Option C is incorrect.

In the lower half of the string tension is lesser, so, speed is lesser, and it will take more time to travel through the lower half compared to upper half.

∴ Option D is incorrect.

59. A long solenoid having 1000 turns per meter carries a current of 1A. It has a soft iron core of $\mu_r = 1000$. The core

is heated beyond the Curie temeperature (T_c). Then

A. The H field in the solenoid is nearly unchanged but the B field decreases drastically.
B. The H and B fields in the solenoid are nearly unchanged.
C. The magnetization in the core reverses direction.
D. The magnetization in the core diminished by a factor of about 10^8

Answer (A)

Sol.
$n = 1000$
$\mu_r = 1000$
$H = nI \rightarrow Remain\ unchanged$
$B = \mu_0(H + I)$

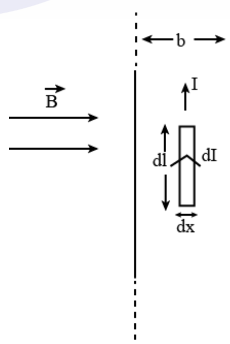
Above curie temperature ferromagnetic substance becomes paramagnetic, *I* decrease by large amount so, B decreases drastically but magnetization does not reverse the direction.

60. A thin infinitely long metal sheet of appreciable finite width b carrying current I (distributed uniformly through out of its cross section) parallel to its length is placed in an external magnetic field B_e Parallel to its plane and perpendicular to the direction of current

A. The thin metal sheet experiences a mechanical pressure $P = \frac{IB_e}{b}$ perpendicular to its face.
B. The direction of the pressure does not change if the direction of current is reversed.
C. In case the external magnetic field B_e is switched off, a magnetic field $B = \frac{\mu_0 I}{2b}$ is observed parallel to the plane of the sheet but perpendicular to the direction currect.
D. The magnetic field produced in part (c) $B = \frac{2\mu_0 I}{b}$

Sol.

Answer (A, C)



$dI = \frac{I}{b}dx$

$d\vec{F} = (dI)\vec{dl} \times \vec{B}_e$

$d\vec{F} = \frac{I}{b}dx dl(\hat{l}) \times B_e(\hat{i})$
$d\vec{F} = \frac{I}{b}B_e(dl)(dx)(-\hat{k})$
$P = \frac{dF}{dA} = \frac{\frac{I}{b}B_e(dl)(dx)}{(dl)(dx)}$
$P = \frac{I}{b}B_e$

If the direction of current is reversed, direction of force is reversed and so, the direction of pressure.

Magnetic field due to infinite plane sheet (near the sheet) is

$$B = \frac{\mu_0 K}{2} = \frac{\mu_0 I}{2b}$$